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* Apply Apriori Algoritham

(1)

TID	items
100	1 3 4
200	2 3 5
300	1 2 3 5
400	2 5

scan D →

Ans :- C_1

itemset	minsup
{1}	2
{2}	3
{3}	3
{4}	1 X
{5}	3

L_1

items	support
{1}	2
{2}	3
{3}	3
{5}	3

frequent - 2

C_2

items	support
{1,2}	1 X
{1,3}	2
{1,5}	1 X
{2,3}	2
{2,5}	3
{3,5}	2

L_2

items	support
{1,3}	2
{2,3}	2
{2,5}	3
{3,5}	2

frequent - 3

C_3

item	support
{1,2,3}	1 X
{1,3,5}	1 X
{2,3,5}	2

L_3

item	support
{2,3,5}	2

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=> Rule Generation

Association Rule	support	confidence	%
2^3 → 5	2	2/2 = 1	100%
3^5 → 2	2	2/2 = 1	100%
2^5 → 3	2	2/3 = 0.66	66%
2 → 3^5	2	2/3 = 0.66	66%
3 → 2^5	2	2/3 = 0.66	66%
5 → 2^3	2	2/3 = 0.66	66%

Strong association → 2^3 → 5
3^5 → 2

(2)

Min-sup = 3

TID	Items
1	Bread, milk
2	Bread, diaper, beer, eggs
3	milk, diaper, beer, cola
4	milk, diaper, beer, cola
5	bread, milk, diaper, cola

frequent 1

C1

Items	Support
{Bread}	3
{diaper}	4
{beer}	3
{eggs}	1 ✗
{cola}	3
{milk}	4

L1

items	support
{bread}	3
{diaper}	4
{beer}	3
{cola}	3
{milk}	4

frequent - 2

C_2 Items	support
{bread, diaper}	2 ✗
{bread, beer}	1 ✗
{bread, cola}	1 ✗
{bread, milk}	2 ✗
{diaper, beer}	3
{diaper, cola}	3
{diaper, milk}	3
{beer, cola}	2 ✗
{beer, milk}	2 ✗
{cola, milk}	3

L_2 Items	support
{diaper, beer}	3
{diaper, cola}	3
{diaper, milk}	3
{cola, milk}	3

C_3 Items	support
{diaper, beer, cola}	2 ✗
{diaper, beer, milk}	2 ✗
{diaper, cola, milk}	3

L_3 Items	support
{diaper, cola, milk}	3

→ Rule Generation

Association Rule	Support	confi.	%
diaper → cola ^ milk	2	3/4 = 0.75	75%
cola → diaper ^ milk	3	3/3 = 1	100%
milk → diaper ^ cola	3	3/4 = 0.75	75%
diaper ^ cola → milk	3	3/3 = 1	100%
diaper ^ milk → cola	3	3/3 = 1	100%
cola ^ milk → diaper	3	3/3 = 1	100%

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FP - Growth Exemple

TID	Items
1	E K M N O Y
2	D E K N O Y
3	A E K M
4	C K M U Y
5	C E I K O

→ step 1 : freq Item set min support ≥ 3

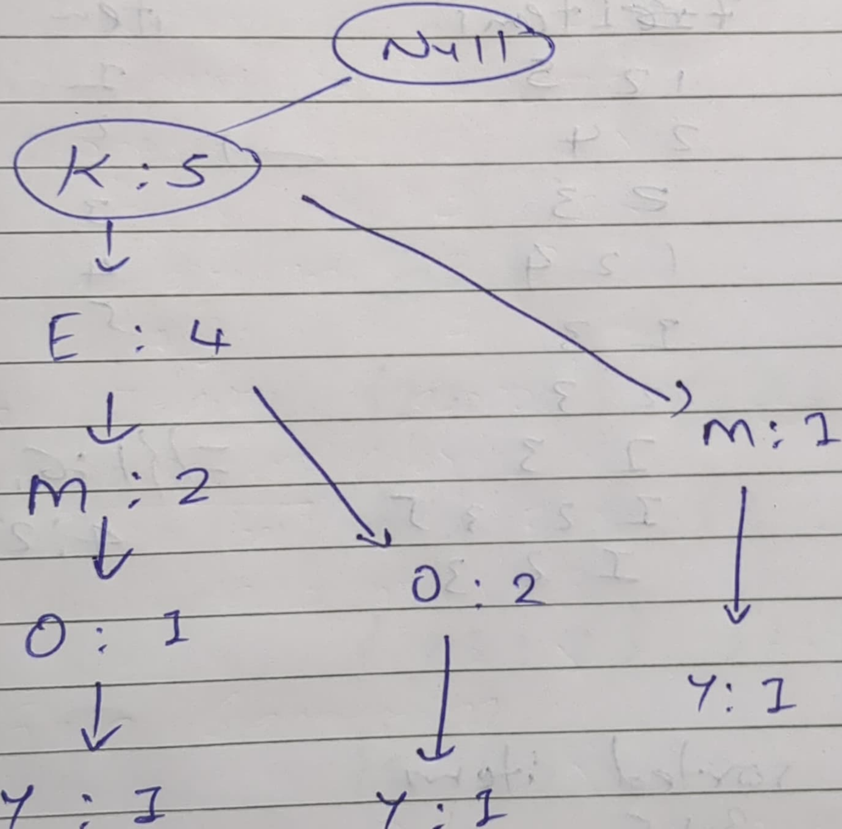
item	frequency	
A	1	×
C	2	×
D	1	×
E	4	✓
I	1	×
K	5	✓
M	3	✓
N	2	×
O	3	✓
U	1	×
Y	3	✓

→ step 2 : transaction with item sorted based on frequency, and ignoring the infrequent items
{K:5, E:4, O:3, M:3, Y:3}

~~III~~

TID	sorted Items
1	K E M O Y
2	K E O Y
3	K E M
4	K M Y
5	K E O

a) Conditional pattern Base



item	Conditional pattern Base	Fptree
Y	{KEMO: 1} {KEO: 1} {KM: 1}	{K: 3}
O	{KEM: 1} {KE: 2}	{K: 3, E: 3}
M	{KE: 2} {K: 1}	{K: 3}
E	{K: 4}	{K: 4}
K	—	

frequent pattern Gen

{K, Y : 3}

{K, O : 3} {E : O : 3} {K, E ; O : 3}

{K, M : 3}

{K. E : 4}

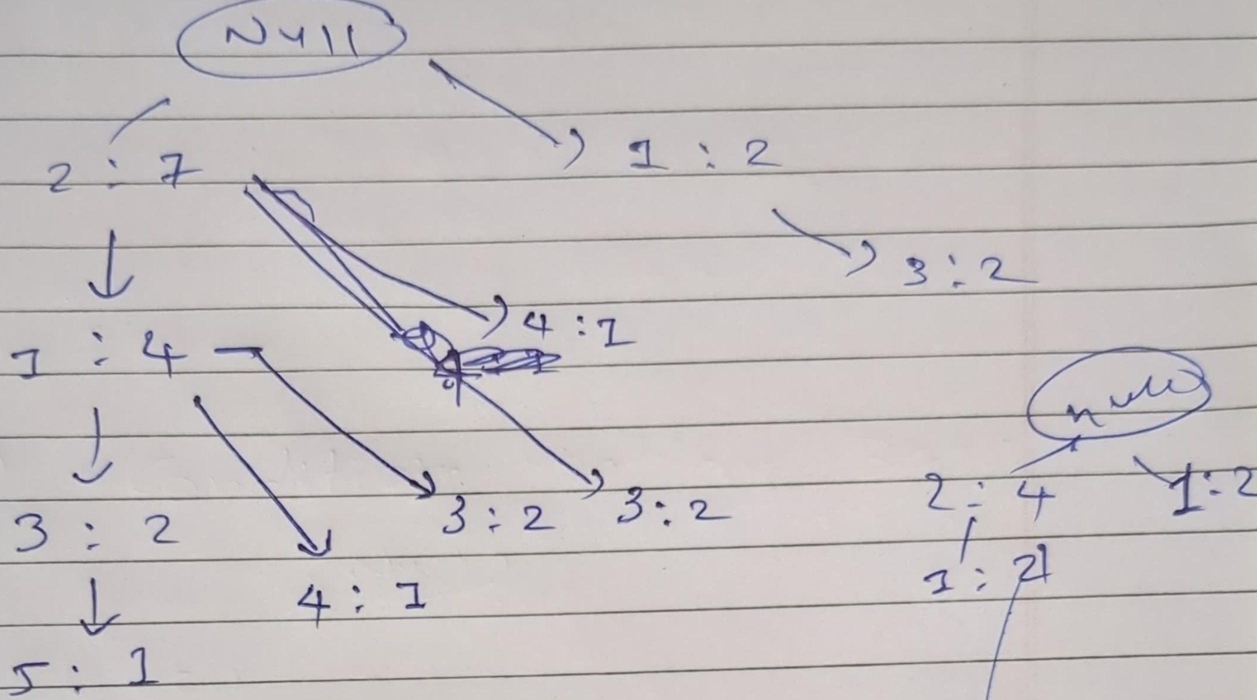
Ex : 2

IID	~~F~~ Item		item	freq
1	1 2 5		1	6
2	2 4		2	7
3	2 3		3	6
4	1 2 4		4	2
5	1 3		5	2
6	2 3			
7	1 3			
8	1 2 3 5			
9	1 2 3			

=> {1 : 6, 2 : 7, 3 : 6, 4 : 2, 5 : 2

=> sorted

IID	sorted item
1	2 1 5
2	2 4
3	2 3
4	2 1 4
5	1 3
6	2 3
7	1 3
8	2 1 3 5
9	2 1 3



item	condition	FP
5	{21 : 1}, {213 : 1}	{12 : 2}
4	{21 : 1}, {2 : 1}	{2 : 2}
3	{21 : 2}, {1 : 2}, {2 : 2}	{2 : 4} {1 : 2}
1	{2 : 4}	{2 : 4}

free.

{2, 5 : 2}, {1, 5 : 2}, {2, 1, 5 : 2}

{2, 4 : 2}

{2, 3 : 4}, {1, 3 : 2}

{2, 1 : 4}